Track 6

Problem Statement

Development of a Single Analytics Platform that Analyzes Telecom CDR/IPDR Data, Bank Statements, and Social Media Activity to Detect Anomalies, Uncover Patterns, and Generate Actionable Insights Across Digital Footprints.

Explanation

Modern criminal and fraud investigations increasingly depend on evidence scattered across heterogeneous digital sources — telecom Call Detail Records (CDR) and Internet Protocol Detail Records (IPDR), bank account statements and transaction histories, and social media activity logs. Each of these data sources is typically analyzed in isolation using separate tools, requiring investigators to manually cross-reference timestamps, identities, locations, and financial flows to reconstruct a coherent picture of a suspect's activities. This fragmented, manual approach is time-consuming, error-prone, and struggles to scale with the growing volume and complexity of digital evidence in contemporary investigations.

There is a pressing need for a unified investigative analytics platform capable of ingesting, correlating, and visualizing data across telecom, banking, and social media domains. Such a platform would allow investigators to detect anomalies (e.g., unusual call patterns, suspicious fund transfers, coordinated social media activity), uncover hidden relationships between entities, reconstruct digital footprints and timelines, and generate actionable, evidence-backed intelligence — all while operating within applicable legal, ethical, and data-privacy frameworks.

Objectives

- To provide investigators with a single platform for ingesting and analyzing telecom CDR/IPDR records, bank statements, and social media activity data.
- To detect anomalies and suspicious patterns such as irregular call/data usage, unusual transaction behavior, or coordinated online activity.
- To correlate identifiers (phone numbers, IMEI/IMSI, bank accounts, IP addresses, social media handles) across data sources to uncover hidden relationships and networks.
- To reconstruct chronological digital footprints and movement/activity timelines of persons of interest.
- To improve the speed, accuracy, and consistency of multi-source digital investigations through automation.
- To generate structured, actionable intelligence reports that support evidence-based decision-making.

- To ensure the platform adheres to applicable legal, ethical, and data-privacy safeguards, including access controls and audit trails.

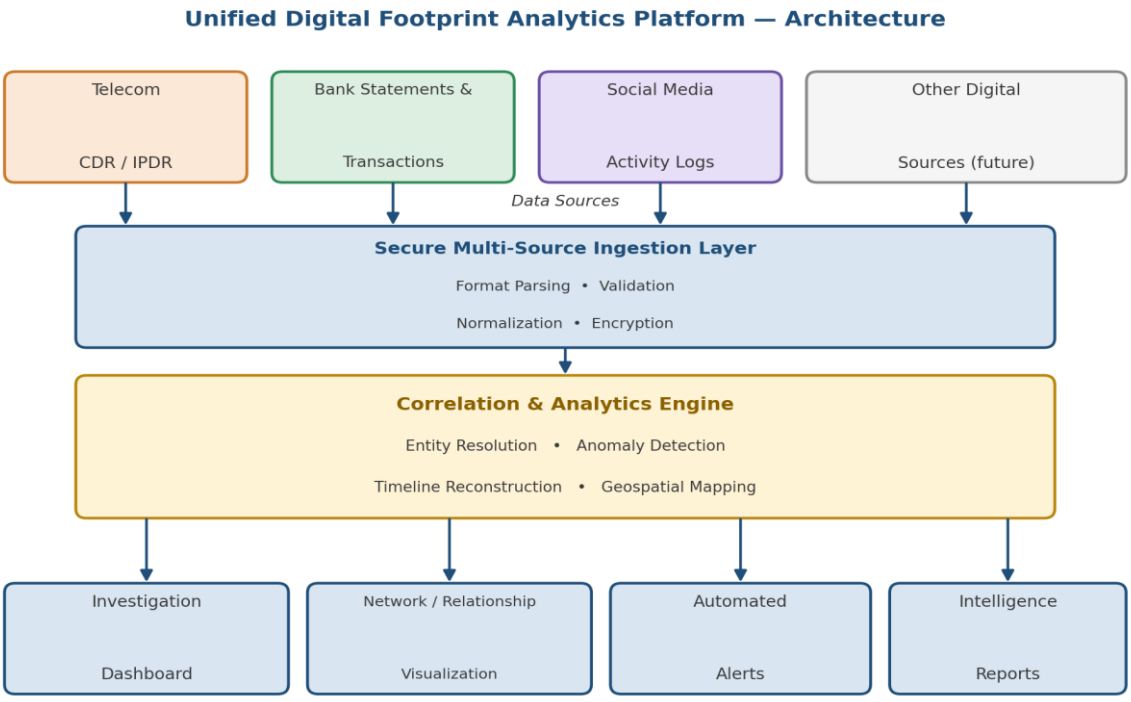


Figure 1: Illustrative high-level architecture of the proposed unified analytics platform, from multi-source ingestion through correlation, analytics, and investigator-facing outputs.

Expected Features

The proposed platform may include the following capabilities:

1. Multi-Source Data Ingestion

Enable structured ingestion of telecom CDR/IPDR files, bank statements (various formats such as CSV, PDF, and standard banking exports), and social media activity data, with support for common formats used by law enforcement and financial institutions.

2. Entity Resolution and Correlation

Identify and link common entities — phone numbers, IMEI/IMSI numbers, bank account numbers, IP addresses, email addresses, and social media handles — across the three data domains to reveal relationships between individuals, accounts, and devices.

3. Anomaly and Pattern Detection

Automatically flag unusual calling patterns, abnormal transaction volumes or frequencies, rapid fund transfers between accounts, sudden changes in device/location usage, and coordinated or bot-like social media behavior.

4. Timeline and Digital Footprint Reconstruction

Reconstruct chronological sequences of calls, transactions, location changes, and social media posts to visualize an individual's or network's activity over a given period.

5. Interactive Investigation Dashboard

Provide a centralized dashboard for investigators to search records, monitor alerts, visualize trends across data sources, and manage ongoing case files.

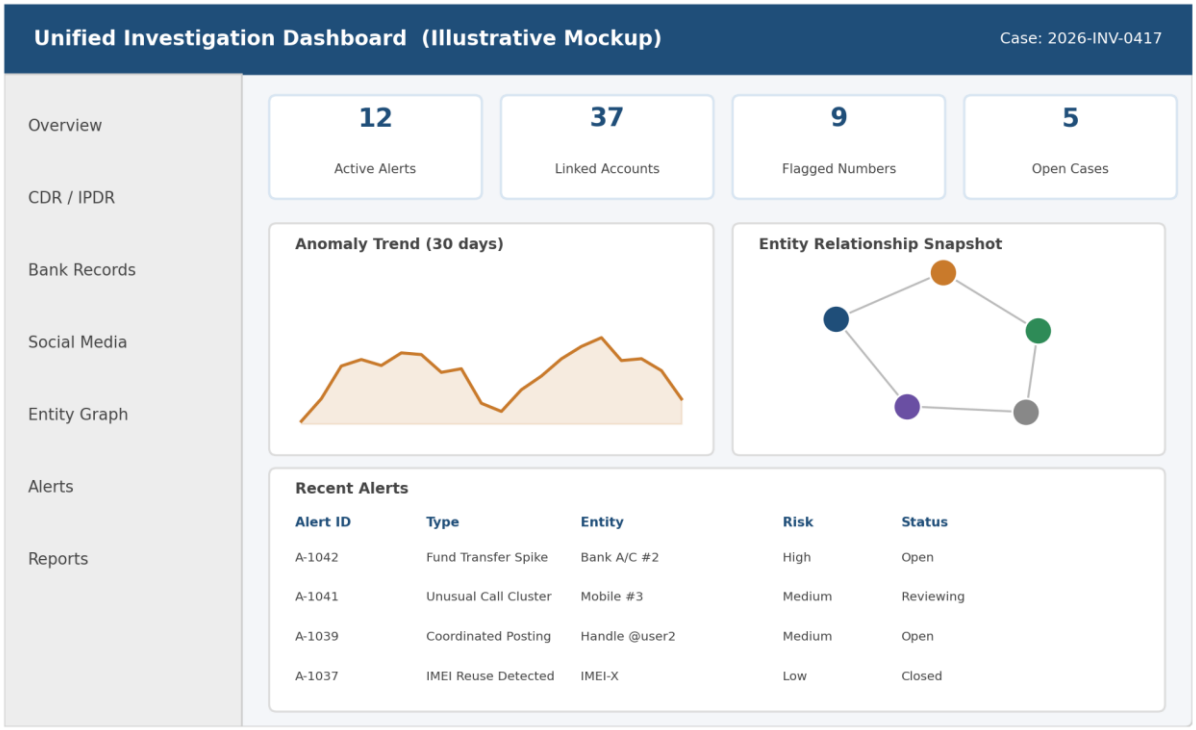


Figure 3: Illustrative mockup of an investigator-facing dashboard, showing KPI summaries, anomaly trends, a relationship snapshot, and a recent-alerts table (conceptual design, not a functional screenshot).

6. Network and Relationship Visualization

Display graph-based visualizations of relationships between suspects, financial accounts, devices, and communication contacts to simplify the analysis of complex networks.

7. Geospatial Analysis

Map cell-tower/IPDR-derived locations and transaction geolocations to support movement analysis and identify overlaps between individuals of interest.

8. Automated Alert Generation

Notify investigators when predefined risk thresholds or suspicious cross-domain patterns are detected, such as a call shortly followed by a large fund transfer and related social media activity.

9. Advanced Search and Case Management

Allow investigators to search using identifiers such as phone numbers, account numbers, aliases, or IP addresses, and to maintain organized, retrievable case histories.

10. Reporting and Evidence Management

Generate structured, court-ready intelligence reports, maintain chain-of-custody logs, and support secure documentation and export of digital evidence.

11. Security and Access Control

Implement secure authentication, role-based access control, end-to-end data encryption, and comprehensive audit logging to protect sensitive investigative data.

12. Scalability and Extensibility

Support continuous ingestion and analysis of large, heterogeneous datasets, with the flexibility to integrate additional data sources (e.g., email metadata, travel records) in the future.

Tech Stack

Participants are encouraged to justify their technology choices based on scalability, security, maintainability, performance, and suitability for the proposed solution, rather than adopting any predefined technology stack.

Resources:

1) Real-World Operation (Delhi Police cyber-fraud/mule-account bust):
https://www.newsgram.com/india/2026/05/05/five-arrested-as-delhi-police-bust-interstate-cyber-fraud-racket

2) Official Initiative (I4C-RBIH MoU on AI-based mule-account detection):
https://ddnews.gov.in/en/i4c-rbih-join-hands-to-use-ai-against-mule-accounts-and-cyber-frauds/

3) National Cybercrime Reporting Portal (I4C, Ministry of Home Affairs):
https://services.india.gov.in/service/detail/report-financial-fraud-through-the-national-cyber-crime-reporting-portal

4) CDR and IPDR Analysis in Indian Criminal Investigations:
https://securiumsolutions.com/cdr-ipdr-analysis-how-call-detail-records-are-used-to-solve-crimes-in-india/